A01 AIR QUALITY | P

Intent : Provide a basic level of indoor air quality that contributes to the health and well-being of building users.

Summary : This WELL feature requires projects to provide acceptable air quality levels, as determined by public health authorities.

Issue : Exposure to air pollutants, such as Volatile Organic Compounds (VOCs), ozone, particulate matter, carbon monoxide and others has been shown to increase the risk of respiratory and cardiovascular diseases, in addition to causing thousands of cancer deaths annually.1 Inhaling pollutants presents symptoms including headaches, dry throat, eye irritation and runny nose that may later develop into extreme health outcomes, such as asthma attacks and cancer.2–4 In addition, radon exposure is the second leading cause of lung cancer, after tobacco use.5 Therefore, it is important to define indoor air quality levels that minimize risk to human health.

Solutions : The World Health Organization (WHO) and other regulatory bodies, such as the U.S. Environmental Protection Agency (EPA) identify a list of "criteria" air pollutants. They have established permissible levels for such criteria pollutants based on epidemiological studies that show the relationships between concentrations of these pollutants, duration of exposure and health risks. Achieving the goal of clean indoor air as defined by permissible levels, requires the joint efforts of both professionals and building users in the implementation of adequate approaches. Indoor air quality can be properly managed through different features listed in the WELL Air concept, including source control strategies, passive and active building design and operation strategies and human behavior interventions. Effective mechanical ventilation is particularly effective at bringing radon below acceptable thresholds.6,7

PART 1 MEET THRESHOLDS FOR PARTICULATE MATTER

For All Spaces except Commercial Kitchen Spaces & Industrial:

Option 1: Acceptable thresholds

The following thresholds are met in occupiable spaces:

a. $PM_{2.5}$: 15 μg/m^3 or lower.[8]

b.

PM_{10}: 50 μg/m^3 or lower.[9]

OR

Option 2: Modified thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Gold regardless of total points achieved.

For buildings where the annual average outdoor PM2.5 level is 35 μg/m^3 or higher, the following thresholds are met:

a. $PM_{2.5}$: 25 μg/m^3 or lower.[10]

b. PM_{10}: 50 μg/m^3 or lower.[10]

OR

Option 3: Dynamic thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For buildings where the annual average outdoor PM2.5 level is 35 μg/m^3 or higher, the following thresholds are met:

a. $PM_{2.5}$ less than or equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

b. PM_{10} less than or equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

For Commercial Kitchen Spaces & Industrial:

Option 1: Acceptable thresholds

The following threshold is met:

a. $PM_{2.5}$: 35 μg/m^3 or lower.[8]

OR

Option 2: Dynamic thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For buildings where the annual average outdoor PM2.5 level is 35 μg/m^3 or higher, the following thresholds are met:

a. $PM_{2.5}$ equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

b. PM_{10} equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

The World Health Organization's Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET THRESHOLDS FOR ORGANIC GASES

For All Spaces:

Option 1: Laboratory-based VOC tests

The following thresholds are met in occupiable spaces:

a. Benzene (CAS 71-43-2): 10 μg/m^3 or lower.[11]

b. Formaldehyde (CAS 50-00-0): 50 μg/m^3 or lower.[12]

c. Toluene (CAS 108-88-3): 300 μg/m^3 or lower.[13]

OR

Option 2: TVOC continuous monitoring

The following threshold is met in occupiable spaces:

a. Total VOC: 500 μg/m^3 or lower.

> **Note** : Projects undergoing recertification which were previously awarded Feature A08 must consider all data collected since the previous (re)certification.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 3 MEET THRESHOLDS FOR INORGANIC GASES

For All Spaces except Commercial Kitchen Spaces & Industrial:

The following thresholds are met in occupiable spaces:

a. Carbon monoxide: 10 mg/m^3 [9 ppm] or lower.[8]

b. Ozone: 100 μg/m^3 [51 ppb] or lower.[10]

For Commercial Kitchen Spaces & Industrial:

The following thresholds are met:

a. Carbon monoxide: 34 mg/m^3 [30 ppm] or lower.[14]

b. Ozone: 100 μg/m^3 [51 ppb] or lower.[10]

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 4 MEET THRESHOLDS FOR RADON

For All Spaces:

Option 1: Radon testing

The following threshold is met on the lowest regularly occupiable floor:

a. Radon: 0.15 Bq/L [4 pCi/L] or lower, as tested by a professional demonstrated not to have a conflict of interest with the WELL project. One test is conducted per 2300 m^2 of regularly occupied space.

OR

Option 2: Mechanical ventilation

For regularly occupied spaces at or below grade, the following requirement is met:

a. All regularly occupied spaces at or below grade meet Feature A03, Part 1, Option 1.

OR

Option 3: Above-grade

One of the following requirements are met in regularly occupied spaces:

a. Is completely located on or above the third floor of the building.

b. Is constructed with raised-pier foundations (e.g., without a solid perimeter wall) and all mechanical equipment is elevated off the ground.

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 5 MEASURE AIR PARAMETERS

For All Spaces except Dwelling Units:

The following pollutants are monitored in occupiable spaces at intervals no longer than once per year, and the results are submitted annually through the WELL digital platform. To determine the required sample locations and quantity, refer to the WELL Performance Verification Guidebook.

a. PM2.5
b. PM10.
c. One of the following:
    1. Total VOC.
    2. Benzene, Formaldehyde, Toluene.
d. Carbon Monoxide.
e. Ozone.

> **Note :** Projects are not required to follow the device requirements or test methods described in the Performance Verification Guidebook.

REFERENCES

1. Franklin BA, Brook R, Arden Pope C. Air pollution and cardiovascular disease. Curr Probl Cardiol. 2015;40(5):207-238. doi:10.1016/j.cpcardiol.2015.01.003
2. Joshi S. The sick building syndrome. Indian J Occup Environ Med. 2008;12(2):61. doi:10.4103/0019-5278.43262
3. Selgrade MK, Plopper CG, Gilmour MI, Conolly RB, Foos BSP. Assessing the health effects and risks associated with children’s inhalation exposures - Asthma and allergy. J Toxicol Environ Heal - Part A Curr Issues. 2008;71(3):196-207. doi:10.1080/15287390701597897
4. U.S. Environmental Protection Agency. Indoor Air Pollution: An Introduction for Health Professionals. https://www.epa.gov/indoor-air-quality-iaq/indoor-air-pollution-introduction-health-professionals.
5. World Health Organization. WHO Handbook on Indoor Radon: A Public Health Perspective. Presented at the: 2009. https://www.who.int/ionizing_radiation/env/9789241547673/en/.
6. Yang S, Goyette Pernot J, Hager Jörin C, Niculita-Hirzel H, Perret V, Licina D. Radon Investigation in 650 Energy Efficient Dwellings in Western Switzerland: Impact of Energy Renovation and Building Characteristics. Atmosphere (Basel). 2019;10(12):777. doi:10.3390/atmos10120777
7. Hadlich DE, Grimsrud DT. Radon in Institutional Buildings : The Impacts of Conservation Strategies. Radon Institutional Build Impacts. 1991;(1990).
8. U.S. Environmental Protection Agency. NAAQS Table. https://www.epa.gov/criteria-air-pollutants/naaqs-table.
9. European Commission. Proposal for a directive of the European Parliament and of the Council on ambient air quality and cleaner air for Europe - COM(2005) 447 Final. Off J Eur Union. 2005;0183:1-67.
10. World Health Organization. Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide. Geneva, Switzerland; 2005. http://apps.who.int/iris/bitstream/handle/10665/69477/WHO_SDE_PHE_OEH_06.02_eng.pdf;jsessionid=CD44465F sequence=1.
11. French Agency for Food Environmental and Occupational Health and Safety. ANSES’s List of Indoor Air Quality Guideline Values.; 2018. https://www.anses.fr/fr/system/files/Tableau_VGAI_Juillet2018EN.pdf.
12. Health Canada. Residential Indoor Air Quality Guideline: Formaldehyde.; 2006. https://www.canada.ca/content/dam/canada/health-canada/migration/healthy-canadians/publications/healthy-living-vie-saine/formaldehyde/alt/formaldehyde-eng.pdf.
13. California Office of Environmental Health Hazard Assessment. Acute, 8-hour and Chronic Reference Exposure Level (REL) Summary. https://oehha.ca.gov/air/general-info/oehha-acute-8-hour-and-chronic-reference-exposure-level-rel-summary. Published 2019.
14. Health and Safety Executive. EH40/2005 Workplace Exposure Limits.; 2011. http://www.hse.gov.uk/pUbns/priced/eh40.pdf.